Total No. of Printed Pages:2

SUBJECT CODE NO: - RNP-8009
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I and Sem - II
Examination November / December 2025 (To be held in February-2026)
Engineering Chemistry

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no. 1 is compulsory
3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 **Solve the following questions (any 10)** 20

a) CNTs are mainly used for their excellent
A) Color B) Electrical and mechanical properties C) Toxicity D) Odor

b) Bakelite is a __________ type of polymer

c) Which of the following is a reinforcement material?
A) Bakelite B) Silicon carbide
C) PVC D) PVA

d) Give synthesis of PVA

e) Define cloud point and pour point of lubricating oil

f) Define coolant give its two examples

g) Differentiate between dry corrosion and wet corrosion

h) What is electroplating give its applications

i) Roasting is generally done for
A) Sulphide ores B) Metal alloys
C) Polymers D) Carbon materials

j) Formula to calculate Viscosity index is.......................

k) Explain any two advantages of lithium-ion batteries

l) Draw diagram of fuel cell

SECTION B

Q2 A. Explain Caustic embrittlement 05
B. Give synthesis properties and applications of PVC 05

Q3	A.	Write a note on ion exchange processes	05
	B.	Explain properties and uses of Ethylene glycol as coolant	05
Q4	A.	Writes a note on wet corrosion	05
	B.	Explain Computed Radiography and digital Radiography	05
Q5	A.	Explain carbon -carbon, ceramic matrix composites	05
	B.	Calculate total hardness of water containing following impurities	05

Impurity	Wt	Molecular wt.
CO_2	13.5ppm	44
$CaSO_4$	65.2ppm	136
$Ca(HCO_3)_2$	21ppm	162
$MgCl_2$	10ppm	95
KCl	5ppm	74.5

Q6	A.	Explain composition, properties, and applications of LNG	05
	B.	Kerosene contain 12% hydrogen and its GCV IS 45 mJ/Kg find net calorific value of kerosene	05
Q7	A.	How corrosion is protected by a) Sacrificial anodic protection b) Electroplating	05
	B.	Define a) Laser cladding b) Casting c) Rolling d) Machining e) Sintering	05